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The Palaeomagnetism of Hematites of the Veldurti Area, Cuddapah Basin, India

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Summary

The palaeomagnetism of thirty-five samples from the hematite deposits of Veldurti (15° 36′ N, 78° E) of Precambrian age has been studied. Most samples show upward dips and reverse polarization. The mean direction for thirty samples showing consistent direction of magnetization is 133° E, −37. The inferred pole position for this group is situated at 45° N and 27° W. The magnetic mineral responsible for magnetization of these samples seems to be hematite. The magnetization is very stable with high coercive force. These observations are consistent with a position for India in the northern hemisphere at the time of magnetization of the deposits.

1. Introduction

Studies of palaeomagnetism of Indian rocks have been mostly devoted to the Deccan traps of Cretaceous to Eocene age, and the Rajmahal and Sylhet Trap of Jurassic age (Clegg, Deutsch & Griffith 1965, Deutsch, Radakrishnamurthy & Sahsrabudhe 1959, Clegg, Radakrishnamurthy & Sahsrabudhe 1958). A few Precambrian formations have also been studied. (Athavale *et al.* 1963). Very little palaeomagnetic work has been done in India on massive iron ore formations. Some of the iron ore deposits from other parts of the world which have been studied are: Allard lake deposits by Carmichael (1959, 1961) and Hargraves (1959), iron grit of Sussex by Edwards (1965), hematite deposits from Egremont, England by Du Bois (1962) and from Czechoslovakia by Hanus & Krs (1963). In this paper results of a palaeomagnetic study of hematite deposits of Veldurti area (15° 36′ N, 78° E) in the Cuddapah Basin of India are reported.

2. Geology and sampling

A general description of the ore deposits has been given by Krishnan (1954). The country rock in the vicinity of Veldurti belongs to the Archean and the Cuddapah Systems. Veldurti is situated on granite gneisses of Archean age. Nearby, basal formations of the Cuddapah Systems, Gulcheru Quartizite, Banganpalle Sandstones and Vempalle Sandstones are exposed. The Veldurti hematite deposits occupy a prominent zone of faulting which has been called the Gunny gull fault. The fault zone cuts across the Archean gneisses and basal formations of the Cuddapah System. The iron ore deposits form low hillocks or ridges whose general trend is E–W to ENE–WSW.

The hematite deposits contain varying amounts of ferric oxide and silica which have probably been deposited simultaneously. The ore bodies are lens-like and form veins and stringers roughly parallel to the trend of the fault. The structure of the deposits suggests that they have been formed by fissure filling in the fault zone which extends over a length of several miles from Ganni Ghattu hill in the east to near Veldurti in the west. Since these were formed, the ores have undergone no marked change.

Thirty-five ore samples collected from several mines have been studied. These mines are known locally as Tappa Reddy mine, Joshi mine, Brahmagundam mine and Gubdibba mine. In the text the word 'sample' means a rock sample collected and the word specimen is used for cylinders cored out of these rocks. Four samples have been analysed chemically for contents of FeO, Fe_2O_3, and TiO_2 with the following result:

Table 1

Chemical analysis of four samples of hematites from Veldurti

	Wt%			
Sample No.	FeO	Fe_2O_3	TiO_2	Rest
V_{19}	0·66	95·39	0·10	3·85
V_{22}	0·99	95·82	0·14	3·05
V_4	1·15	88·82	0·04	10·0
T1–12	0·66	92·98	0·02	6·34

These show that the deposits consist predominantly of hematite and contain less than 1% ilmenite. Some of polished sections show minute exsolution lamellae of ilmenite in a mass of hematite under the microscope.

3. Directions of magnetization

Out of thirty-five samples 170 specimens of 1 in. diameter were studied. Measurements were made using an astatic magnetometer with a sensitivity of 2×10^{-6} oe/cm deflection at 4 m light path. Most samples showed negative (up) dips with declination varying between 74° and 210°. Six samples showed consistent positive dips with azimuth varying between 171° and 320°. Histograms of azimuth and inclination are shown in Figs. 1a and 1b.

Directions of north-seeking polarization for samples showing consistent direction of magnetization are plotted on a stereographic projection in Fig. 2. Five samples gave results inconsistent between specimens from the same rock sample with wide variation in dips and declinations. These were rejected.

4. Stability tests

The specimens were stored in the laboratory in random orientations and were remeasured after about 2 months. Most specimens showed no change in direction.

Eleven specimens were subjected to thermal demagnetization tests in a furnace placed below the magnetometer. The furnace is similar to that of Wilson (1962). The specimens were heated to nearly 700 °C and the magnetization determined at intervals of about 60 °C. Within the accuracy of measurements the vectors remained unchanged up to about 600 °C. Magnetization was lost in most cases at about 640–650 °C. Thus it would seem that the mineral responsible for magnetization is nearly pure hematite (Curie point 670 °C). In most cases there was rapid decrease in intensity of magnetization between 500 and 600 °C.

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Twenty specimens were subjected to alternating field demagnetization tests in an apparatus similar to that of Creer (1959). It was found again that the magnetization is very stable with little or no change in direction for most specimens. The specimens with positive dips are as stable as those with negative dips. For few specimens the intensity decreased to about 30% to 40% of its initial value in alternating fields of 800 to 1 000 oe. In ten cases the decrease in intensity was less than 10% up to this field strength. In three specimens there was increase in the intensity of magnetization above about 500 oe. Typical decay curves are shown in Figs. 3 and 4.

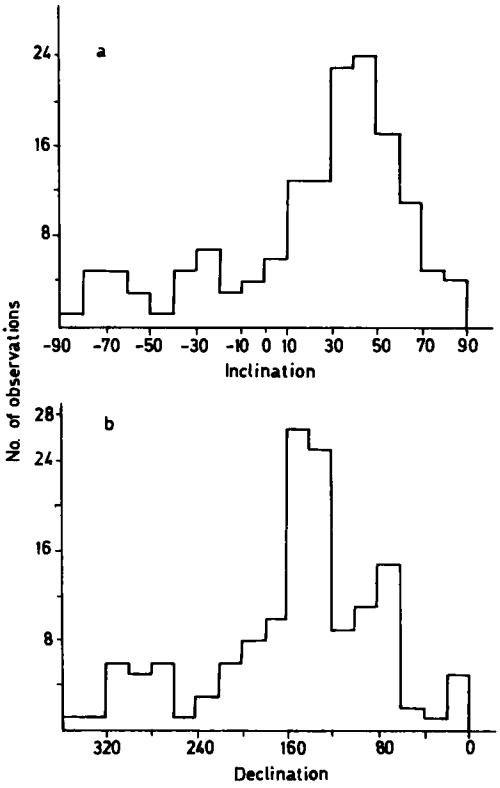


FIG. 1. Histograms of declination and inclination for all specimens.

5. Susceptibility and remanence

A histogram of the intensity of remanent magnetization before a.c. demagnetization of the specimen is plotted in Fig. 5.

For most of the specimens the intensity is below 500×10^{-6} emu/cm^3. Magnetic susceptibility was determined using an a.c. bridge (Fig. 6). The mean value of susceptibility is 117×10^{-6} emu/cm^3, the mean value of I_r is 670×10^{-6} emu/cm^3. Low susceptibility like this suggests that hematite is the main magnetic constituent of the ore. A comparison was made between the intensity of NRM and the magnetic susceptibility. In general, higher values of I_r are associated with higher values of susceptibility. A plot of I_r against susceptibility is shown in Fig. 7.

Intensity of induced magnetization in the Earth's field (H) was estimated putting $H = 0{\cdot}5$ oe. The ratio of the remanent to induced magnetization Q_n, (the Konigsberger ratio) has a large variation, from 1·0 to about 30, with an average of 11·6.

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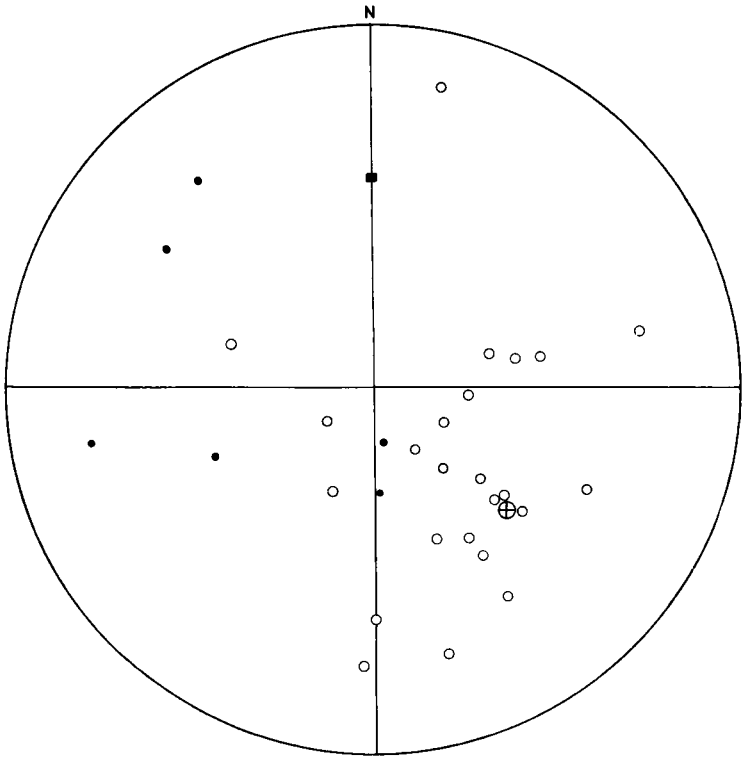


FIG. 2. Stereographic projection of magnetic vectors showing consistent directions of magnetization. Vectors for five samples showing inconsistent directions between specimens were left out. Downward inclinations are plotted as solid circles on the lower hemisphere and upward inclinations as open circles on the upper hemisphere. Up dips, ○; down dips, ●; present dipole field, ■; mean direction, ⊕.

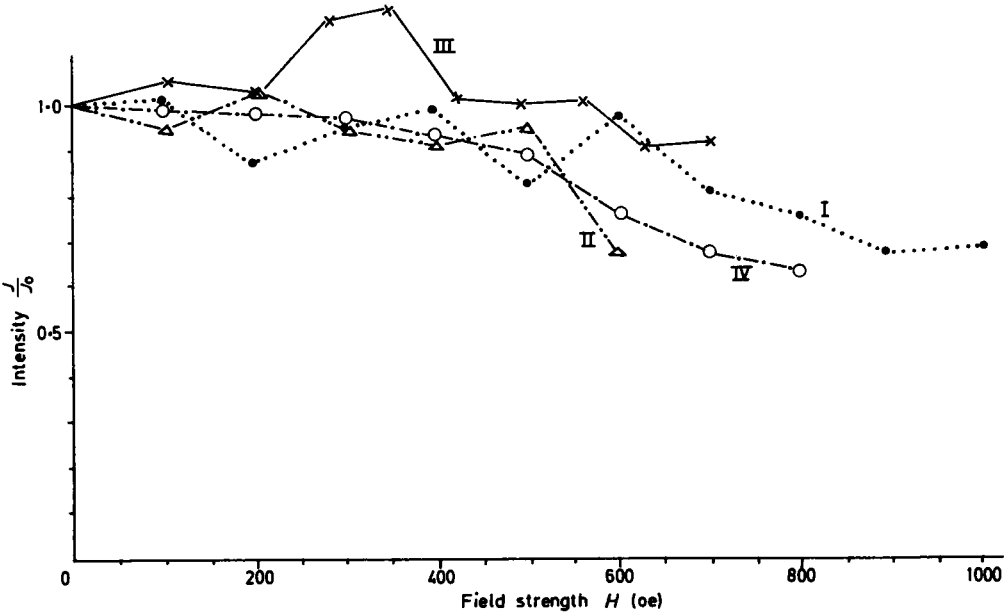


FIG. 3. Normalized a.c. demagnetization curves for three reverse specimens, curves I, II, IV and one normal specimen curve, III.

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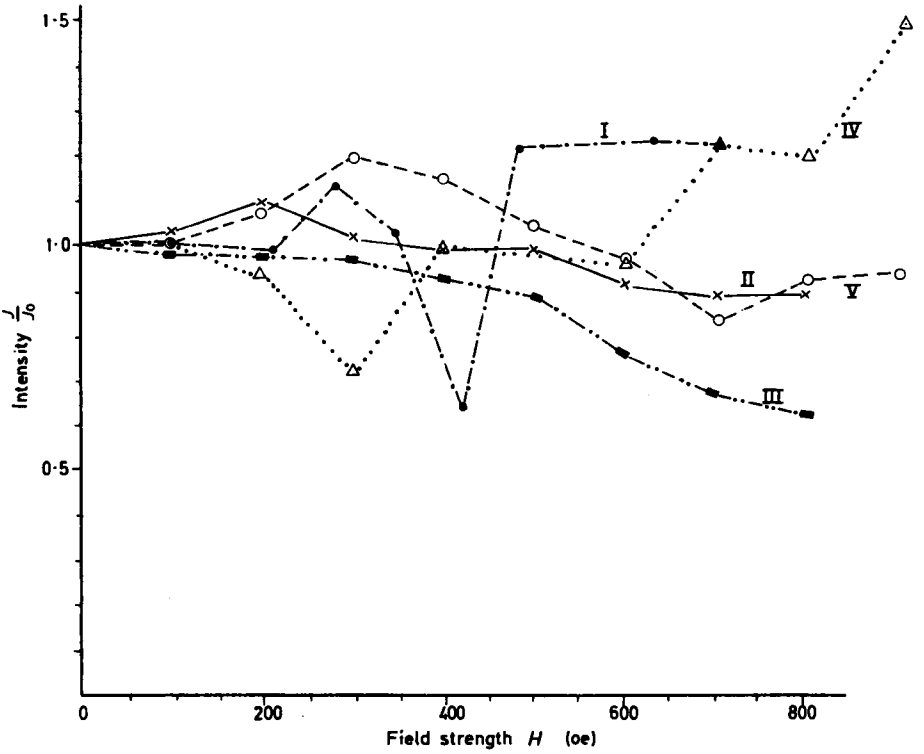


FIG. 4. Normalized a.c. demagnetization curves for four reverse specimens, curves I, II, III, IV and one normal specimen curve, V.

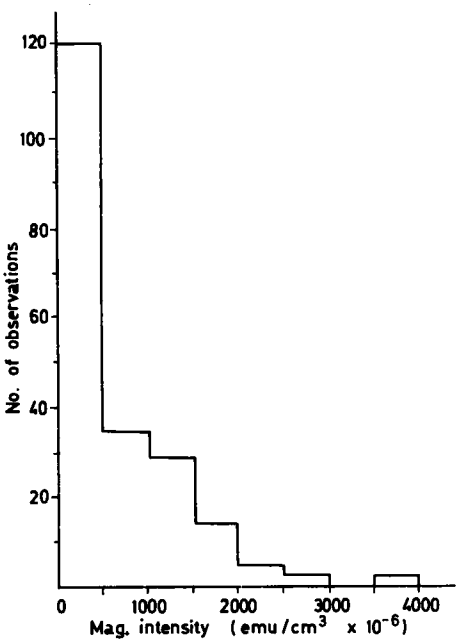


FIG. 5. Histogram of intensity of magnetization for all specimens.

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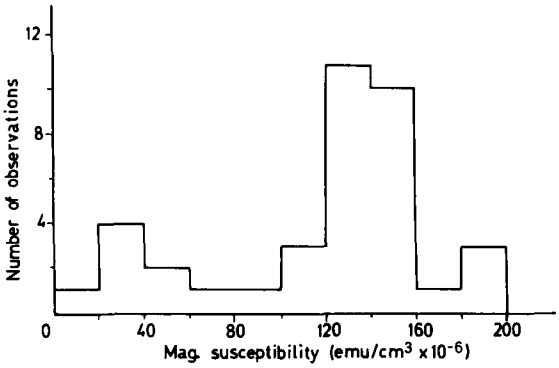


FIG. 6. Histogram of magnetic susceptibility for all samples.

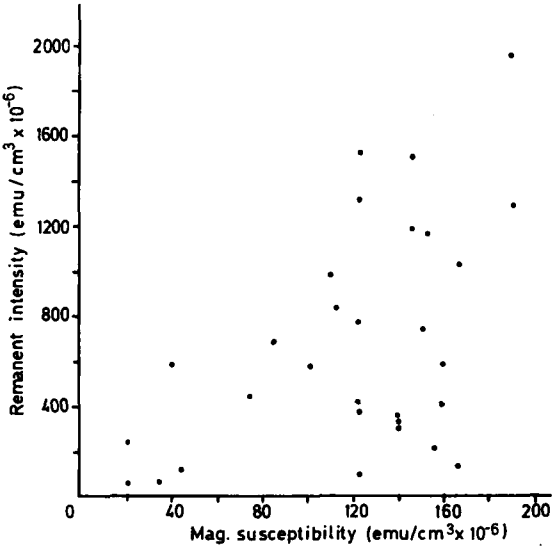


FIG. 7. Remanent intensity against magnetic susceptibility for thirty samples.

6. Analysis and discussion

Results of calculations of resultant mean vector, ancient poleposition and other parameters for thirty rock samples are as given below. Polarity of 'normal' samples was reversed for purposes of calculations.

Mean inclination = −37° (up)

Mean declination = 133°

Rotation ψ = −47

Length of vector sum of all individual vectors, R = 21·7

Fisher's precision parameter, K = 3·5

Radius of 95% circle of confidence, $\alpha 95$ = 14°

Ancient latitude of site = 21° N

Ancient pole position

latitude λp = 45° N

longitude Lp = 27° W.

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Precambrian volcanic rocks of Mundwara Complex in Sirohi District of Rajasthan (25° N, 73° E) were previously studied by Athavale *et al.* (1963). The estimated age of these rocks is approximately 750 m.y. The pole position for these rocks is 42° N, 65° W. This is the nearest pole position for any Precambrian Indian formation to the pole position obtained for Veldurti hematites. Most of the hematites are reversely magnetized while rocks from Mundwara Complex are normally magnetized. The results from these hematites are consistent with the position of India in the northern hemisphere at the time of their magnetization.

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